Date: November 2010 G.C

Name ______________________________ ID.NO.______________________

☛ For the Instructor's use only

Item	Part I (4%)	Part II (4%)	Part III (7%)	Total (15%)
Score				

I. Fill in the blank space with the most appropriate answers [4%, 1 pt each]

[1]. Given two vectors $\vec{A} = 3i - j + 4k$ and $\vec{B} = -5i + 4j + 2k$ then $|\vec{A} + \vec{B}| =$ ______________.

[2]. If $\vec{A} = 2i + 2j + k$, $\vec{B} = -i + 3j + 4k$ and $\vec{C} = i - j - k$, then $\vec{A} \bullet (\vec{B} + \vec{C}) =$ ______________.

[3]. If $\vec{A} = 2i + \lambda j + k$, $\vec{B} = i - 4j + 2k$ are perpendicular, the value of the constant λ is ______________.

~~[4]. If~~ $\cancel{\vec{A}(u) = (u-1)i + (2u+3)j + 4u^2 k}$ ~~then~~ $\cancel{\int_2^3 \vec{A}(u)\,du}$

II. Choose the correct answer and write the letter of your choice on the space provided. [4%, 1 point each]

____ [5]. For any two non zero vectors $\vec{A}$ and $\vec{B}$ if $|\vec{A} + \vec{B}| = |\vec{A}| + |\vec{B}|$, then:

A. $\vec{A}$ and $\vec{B}$ are parallel and in the same direction

B. $\vec{A}$ and $\vec{B}$ are parallel and in opposite direction

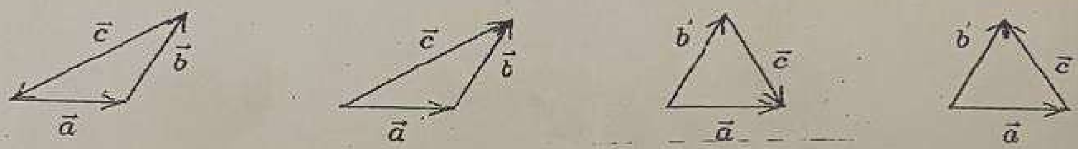
C. The angle between $\vec{A}$ and $\vec{B}$ is $45°$

D. $\vec{A}$ and $\vec{B}$ are perpendicular to each other

____ [6]. Which one of the following is **not true** about two non vanishing vectors $\vec{A}$ and $\vec{B}$?

A. $\vec{A} \bullet \vec{B} = \vec{B} \bullet \vec{A}$ B. $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$ C. $\vec{A} + \vec{B} = \vec{B} + \vec{A}$ D. $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$

____ [7.] The vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ are related by $\vec{c} = \vec{b} - \vec{a}$. Which diagram below illustrates this relationship?



A B C D

____ [8.] A vector of magnitude 20 is added to a vector of magnitude 25. The magnitude of this sum might be:

A. zero B. 3 C. 12 D. 47

III: Solve the following problems by showing all the necessary steps on the space provided [7%]

[9]. Given three vectors $\vec{A} = 3i - j + 4k$ and $\vec{B} = -5i + 4j + 3k$ and $\vec{C} = 4i - 6j - k$, then find:

(a) the magnitude of the resultant vector and the angle that the resultant vector makes with respect to +y-axis.

10. The magnitude and directions with respect to x-axis of two vectors are $\vec{A} = (50, 30^0)$ and $\vec{B} = (25, -60^0)$ find the components of $\vec{A}$ and $\vec{B}$, $\vec{A} + \vec{B}$ and $\vec{A} - \vec{B}$?